Department of Computer Science and Engineering

Course Code:CSE430 **Credits: 1.5**

Course Name: Digital Signal Processing (DSP) **Prerequisite:CSE111, MAT120**

Lab 01

Basics of MATLAB for DSP

I. Lab Overview:

Learn Basic MATLAB programming concepts such as: different window types (i.e., Command Window, Figure Window, and Editor Window), Matrix, and Function.

II. Why MATLAB for DSP course:

Signal processing is essential for a wide range of applications, from data science to real-time embedded systems. MATLAB products make it easy to use signal processing techniques to explore and analyze time and frequency distributed data, and they provide a unified workflow for the development of embedded systems and streaming applications.

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III. Lesson Fit:

There is pre-requisite to this lab: CSE111, MAT120. You should have intensive Programming Knowledge and capability of understand Integral Calculus and Differential Equations.

IV. Acceptance and Evaluation

Performed lab tasks will be evaluated by the Lab Instructor (LI)

a. Short viva will be conducted in each Lab or occasionally to examine your work.
b. You may work in groups but be aware that you will be evaluated individually; hence active participation during the Lab work demonstration is recommended.

c. There will be Lab handout after your work you have to handover it to LI

V. Learning Outcome:

After this Lab, the students will be able to:

a. Understand basic MATLAB codes and solve basic programming problems in MATLAB.

VI. Activity Detail

a. Hour: 1

Getting Started:

I. Have a glance at Books "John G. Proakis, Dimitris Manolakis,"Digital Signal Processing: Principles, Algorithms and Applications", Prentice Hall, 4th edition, 1996.

II. Richard G. Lyons,"Understanding Digital Signal Processing", Prentice Hall, 3rd edition, 2010. "

III. Check \\TSR to see e-book copy and codes, tutorials and useful links

Running IDE and Code

I. Run "MATLAB.EXE from the root Icon which is shown in desktop.

II. Your Lab PC should have MATLAB software installed if not then inform Lab instructor immediately

Get to know about IDE

MATLAB works with three types of windows on your computer screen. These are the

- Command window: The command window is the main window in which you communicate with the MATLAB interpreter. The MATLAB interpreter displays a command >> indicating that it is ready to accept commands from you.
- Figure window: The Figure window only pops up whenever you plot something.
- Editor window: The Editor window is used for writing and editing MATLAB programs (called M-files) and can be invoked in Windows from the pull-down menu after selecting

File | New | M-file. In UNIX, the Editor window pops up when you type in the command window: edit filename ('filename' is the name of the file you want to create).

The Command Window

View the MATLAB introduction by typing

>> intro

at the MATLAB prompt. This short introduction will demonstrate some basic MATLAB commands.

Explore MATLAB's help capability by trying the following:

```
>> help
>> help plot
>> help ops
>> help arith
```

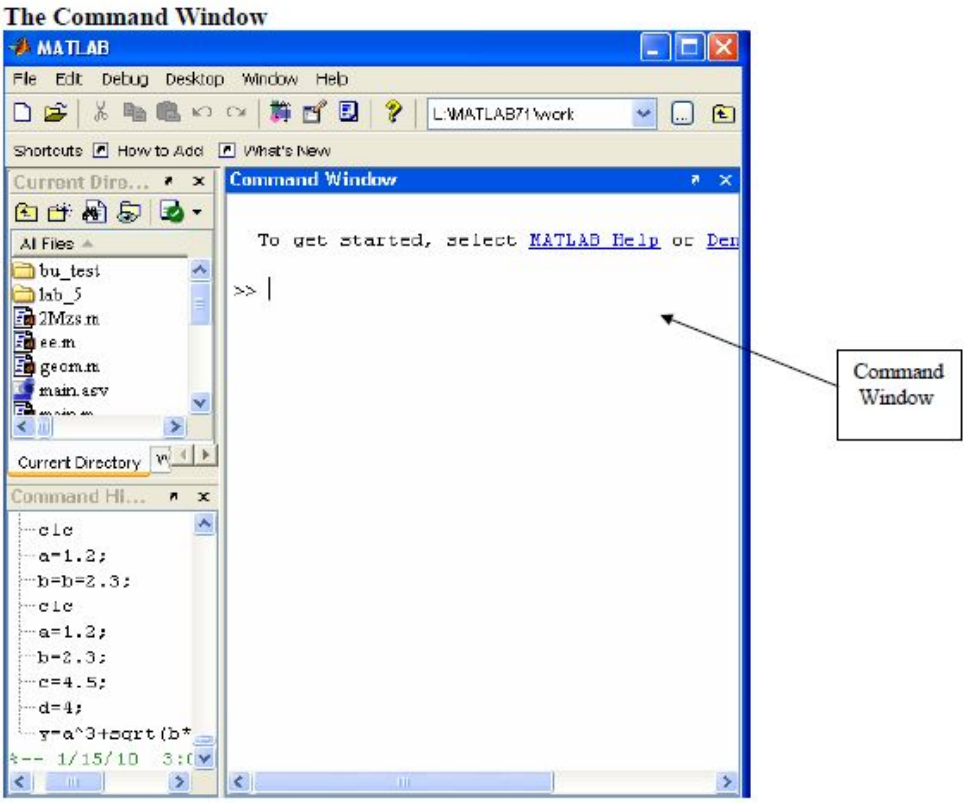


Figure 1: The MATLAB Environment

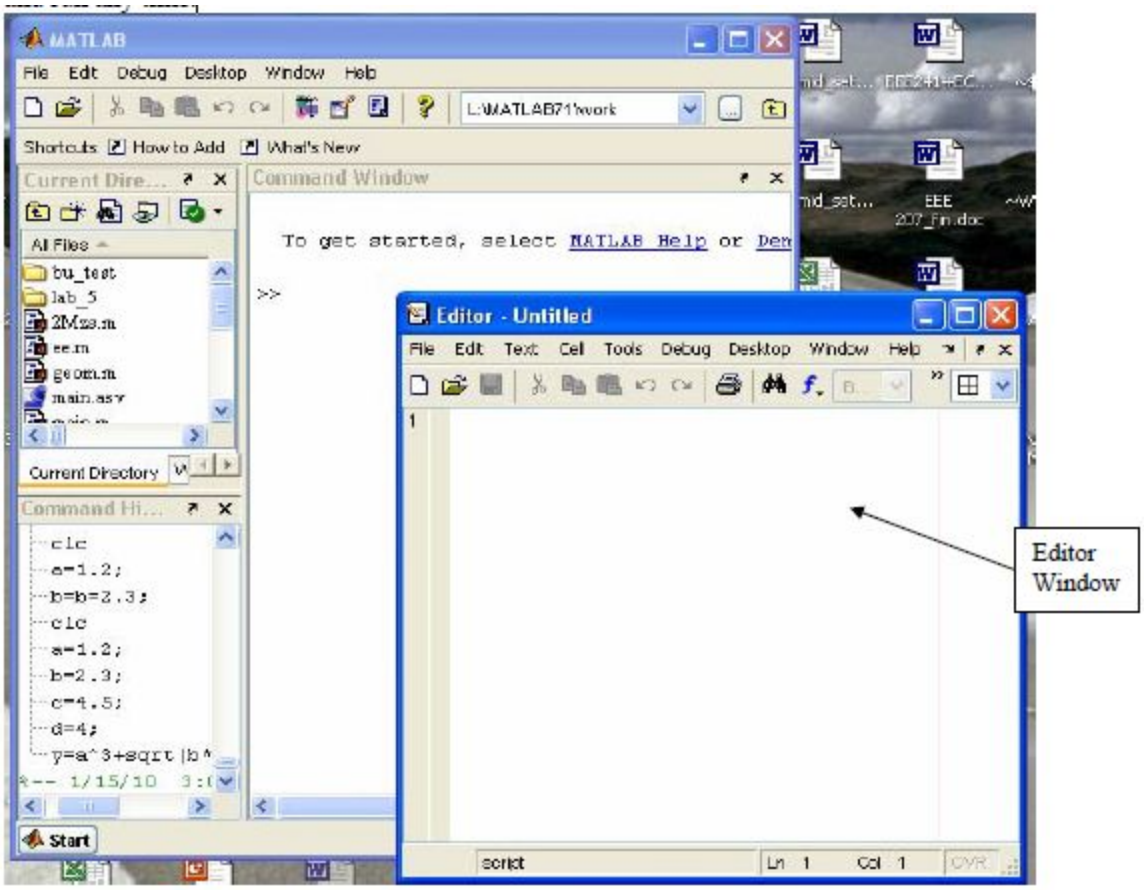
Type demo and explore some of the demos of MATLAB commands.

You can use the command window as a calculator, or you can use it to call other MATLAB programs (M-files).

The Editor Window

Go to **File → New→ M File**

In the editor window you can write any code and save it as a M File. The M file can be saved and run any time.



Suppose you want to evaluate the expression y = $a + b$, where $a = 1.2$ and $b = 2.3$. This can be done in two ways:

Lab 01

- In Command Window

```
Command Window
>> a=1.2;
>> b=2.3;
>> y=a+b

y =

    3.5000
```

- By Creating separate M file

```
Editor - C:\Users\zavid.parvez\lab01task01.m
lab01task01.m
clc
clear all
close all

a=1.2;
b=2.3;
y=a+b
```

Note the semicolon after each variable assignment. If you omit the semicolon, then MATLAB echoes back on the screen the variable value.

There are four different arithmetic operators:

- + addition
- - Subtraction
- * Multiplication
- / Division (for matrices it also means inversion)

Matrices

Every element in Matlab is a matrix. So, for example, the Matlab command
a = [7 3 4] creates a matrix named "a" with dimensions of 1 × 3. To access a specific entry inside the matrix, one uses the index representing the position of the desired entry in the matrix.

For example a[2] corresponds to the number 3 in our previous example.

The variable "a" is stored in what is called the Matlab workspace. The operation
b = a.' stores the transpose of "a" into the vector "b". In this case, "b" is a 3 × 1 vector.

Since each element in Matlab is a matrix, the operation
c = a*b; computes the matrix product of "a" and "b" to generate a scalar value for "c" of 74 =7 × 7 + 3 × 3 + 4 × 4.

Often, you may want to apply an operation to each element of a vector. For example, you many want to square each value of "a". In this case, you may use the following command.

```
c = a.*a
```

The dot before the * tells Matlab that the multiplication should be applied to each corresponding element of "a". Therefore the .* operation is not a matrix operation. The dot convention works with many other Matlab commands such as divide ./, and power .^. An error results if you try to perform element-wise operations on matrices that aren't the same size.

Note also that while the operation a.' performs a transpose on the matrix "a", the operation a' performs a conjugate transpose on "a" (transposes the matrix and conjugates each number in the matrix).

Functions

To create a function call "func", you first create a file called "func.m". The first line of the file must be

```
function output = func(input)
```

where "input" designates the set of input variables, and "output" are your output variables. The rest of the function file then contains the desired operations. All variables

in the function are local; that means the function cannot access Matlab workspace variables that you don't pass as inputs. After the execution of the function, you cannot access internal variables of the function.

Activity List

Task 01: Mark 10 **Time:** 1 hour

Solve the following problems with MATLAB programming language

Question 1. Write a program to convert temperature from Centigrade (C') to Fahrenheit (F')

Far=1.8Cel+32, where *Cel* is 100^0.

Question 2. A vector has 6 numbers. Print the smallest and then largest number,
Hints: A = [10 20 5 3 100];

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